



Research Intake 2026

Day 3

Supervised Learning: Regression

A Beginner's Guide for Students

Understanding AI, Machine Learning, and Deep Learning

Gudsky Research Foundation

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1. Introduction to Supervised Learning

1.1 What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence that enables systems to learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can access data and use it to learn for themselves.

Three Main Types of Machine Learning:

- **Supervised Learning:** Learning with labeled data
- **Unsupervised Learning:** Learning with unlabeled data
- **Reinforcement Learning:** Learning through rewards and penalties

1.2 Supervised Learning Explained

Supervised learning is like having a teacher guide you. The algorithm learns from labeled training data, where each example comes with the correct answer (label).

Key Components:

- **Input (X):** Features or independent variables
- **Output (Y):** Target or dependent variable (label)
- **Function (f):** The mapping we want to learn: $Y = f(X)$

Two Main Categories of Supervised Learning:

1. **Classification:** Predicting discrete categories
 - Email: Spam or Not Spam
 - Image: Cat or Dog
 - Disease: Present or Absent
2. **Regression:** Predicting continuous values
 - House prices
 - Temperature
 - Stock prices
 - Sales forecasting

1.3 Why Regression?

Regression analysis is fundamental in predictive modeling because:

- It quantifies relationships between variables
- It enables forecasting and prediction
- It helps understand cause-and-effect relationships
- It forms the foundation for more complex models

2. Understanding Regression

2.1 What is Regression?

Regression is a statistical method used to model and analyze the relationship between a dependent variable (target) and one or more independent variables (features). The goal is to find the best-fitting function that describes this relationship.

Real-World Analogy: Imagine you're trying to predict how much ice cream a shop will sell based on temperature. As temperature increases, ice cream sales tend to increase. Regression helps us quantify this relationship mathematically.

2.2 Different Names and Terminology

Linear regression has been around since 1805 and has been studied from every possible angle. As a result, it has accumulated many different names and terminology:

Model Types:

- **Simple Linear Regression:** One independent variable
- **Multiple Linear Regression:** Two or more independent variables
- **Polynomial Regression:** Non-linear relationships using polynomial terms
- **Univariate Regression:** Another name for simple linear regression
- **Multivariate Regression:** Multiple dependent variables (less common)

Variable Names (all referring to the same concepts):

- **Independent Variable** = Predictor Variable = Feature = Input = Explanatory Variable = Covariate = X
- **Dependent Variable** = Response Variable = Target Variable = Output = Outcome = Y

An Easy Way to Remember:

- **One** independent variable → **Simple** Linear Regression
- **Multiple** independent variables → **Multiple** Linear Regression

The Main Aim: Behind every regression model is to **predict a value using some features**.

2.3 When to Use Linear Regression

Linear regression is only appropriate when there is a **linear relationship** among variables. How do you check this?

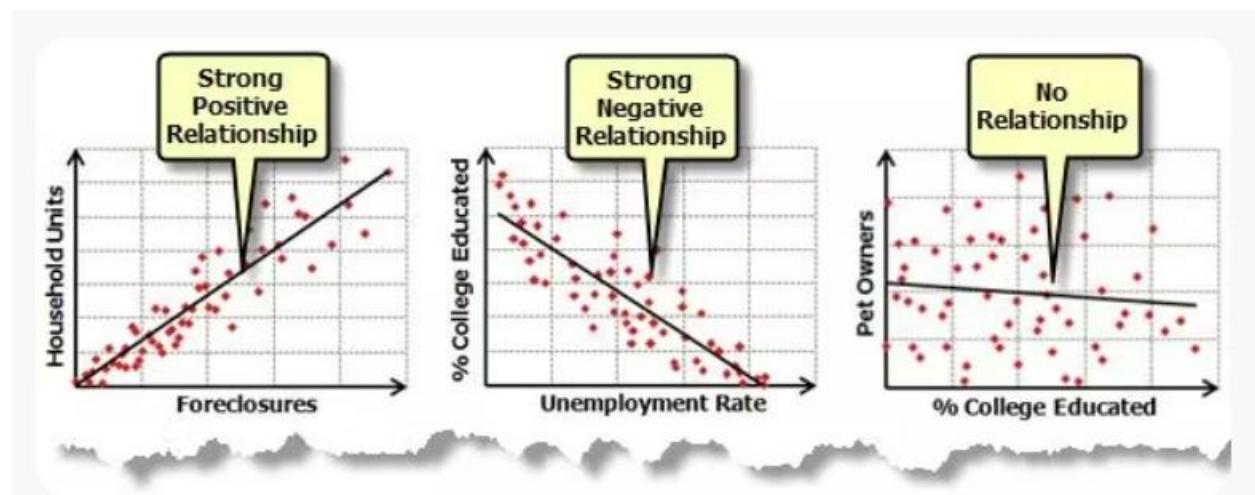
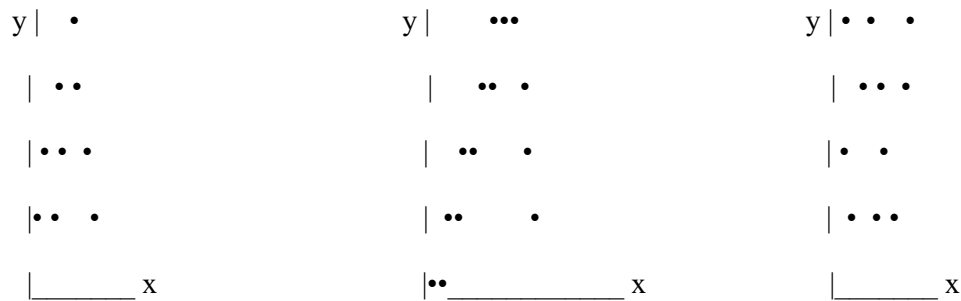
Simple Check: Make a **pair plot** or **scatter plot** between the variables. The graph will show if linear regression is applicable or not.

What to look for:

- **Linear pattern:** Points roughly follow a straight line → Use linear regression
- **Curved pattern:** Points follow a curve → Consider polynomial regression
- **No pattern:** Points are scattered randomly → Linear regression won't work well

Example:

Good for Linear Regression: Not Good (Non-linear): Not Good (No relationship):



2.4 The Regression Line

The regression line (or curve) is the line of best fit that minimizes the distance between the predicted values and actual values.

Mathematical Representation:

$$\hat{y} = f(x)$$

Where:

- $\hat{y}$ (y-hat) = predicted value
- x = input feature

- f = the function we're learning

2.3 Key Concepts

Residuals (Errors): The difference between actual and predicted values:

$$e = y - \hat{y}$$

Best Fit: The line that minimizes the sum of squared residuals (errors).

Assumptions of Linear Regression:

1. **Linearity:** Relationship between X and Y is linear
2. **Independence:** Observations are independent of each other
3. **Homoscedasticity:** Constant variance of residuals
4. **Normality:** Residuals are normally distributed
5. **No multicollinearity:** Features are not highly correlated (for multiple regression)

2.4 Types of Relationships in Linear Regression

Linear regression can reveal different types of relationships between variables. Understanding these relationships is crucial for interpreting your model correctly.

Positive Relationship

When the regression line between two variables moves in the same direction with an **upward slope**, the variables are said to be in a positive relationship. This means that if the value of x (independent variable) is increased, there will be an increase in the dependent variable (y).

Examples:

- Study hours vs. exam scores (more study $\rightarrow$ higher scores)
- Advertising spend vs. sales (more ads $\rightarrow$ more sales)
- Temperature vs. ice cream sales (hotter $\rightarrow$ more sales)

Mathematical Indicator: $\beta_1 > 0$ (positive slope)

Negative Relationship

When the regression line between two variables has a **downward slope**, the variables are said to be in a negative relationship. This shows that if the value of the independent variable (x) is increased, there will be a decrease in the dependent variable (y).

Examples:

- Car age vs. resale value (older $\rightarrow$ less valuable)

- Interest rates vs. home sales (higher rates → fewer sales)
- Drug dosage vs. symptoms (more medication → fewer symptoms)

Mathematical Indicator: $\beta_1 < 0$ (negative slope)

No Relationship

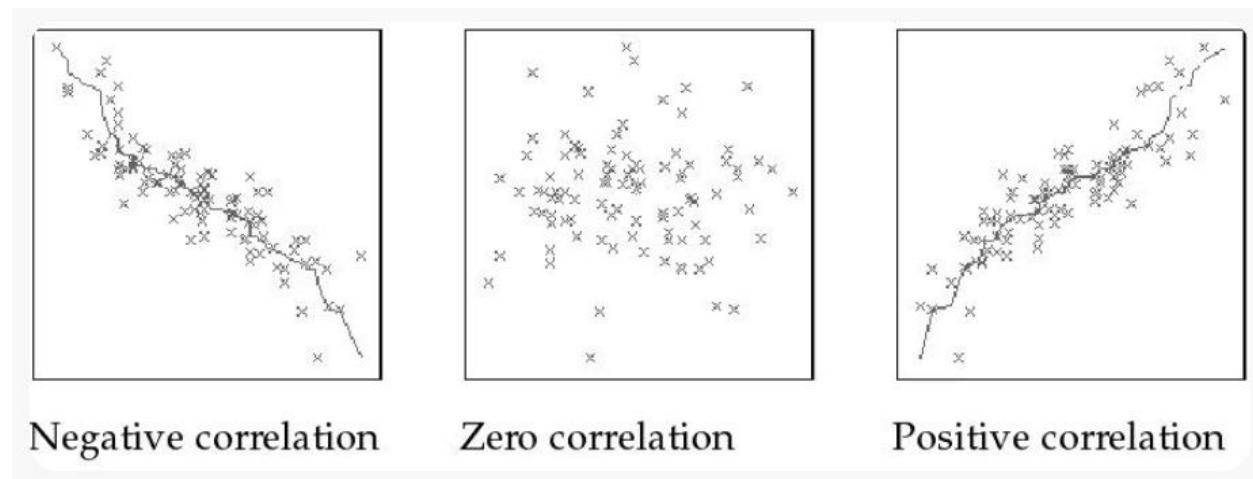
If the best fit line is **flat** (not sloped), it's assumed that there is no relationship among the variables. There will be no change in the dependent variable (y) by increasing or decreasing the independent variable (x) value.

Examples:

- Shoe size vs. intelligence (no relationship)
- Day of the week vs. stock market returns (generally no relationship)

Mathematical Indicator: $\beta_1 \approx 0$ (slope near zero)

Visual Representation:



Measuring Relationship Strength

Covariance: Shows the **direction** of the relationship between X and Y but it doesn't say how strong the relationship is.

$$\text{Cov}(X,Y) = \sum (x_i - \bar{x})(y_i - \bar{y}) / (n-1)$$

Interpretation:

- Positive covariance: Variables move together
- Negative covariance: Variables move in opposite directions
- Zero covariance: No linear relationship

Limitation: The magnitude depends on the scale of variables, making comparison difficult.

Correlation: A statistical measure that shows **both the direction and the strength** of the relationship (how positively or negatively the variables are correlated).

$$r = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$$

Range: $-1 \leq r \leq +1$

Interpretation:

- **r = +1:** Perfect positive correlation (all points on line, positive slope)
- **r > 0.7:** Strong positive correlation
- **r = 0.5:** Moderate positive correlation
- **r ≈ 0:** No linear correlation
- **r = -0.5:** Moderate negative correlation
- **r < -0.7:** Strong negative correlation
- **r = -1:** Perfect negative correlation (all points on line, negative slope)

Perfect Correlation: It's called a perfect correlation if all points fall exactly on the best fit line - which is very unlikely in real-world data due to measurement error and natural variation.

2.5 Cost Function

The cost function measures how well our model performs. For regression, we commonly use Mean Squared Error (MSE):

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Where:

- n = number of observations
- y_i = actual value
- $\hat{y}_i$ = predicted value

Goal: Minimize the cost function to find the best parameters.

3. Simple Linear Regression

3.1 Definition and Formula

Simple Linear Regression models the relationship between ONE independent variable (X) and a dependent variable (Y) using a straight line.

The Equation:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Or for predictions:

$$\hat{y} = \beta_0 + \beta_1 x$$

Where:

- **y:** Actual value of dependent variable
- **$\hat{y}$:** Predicted value
- **β_0 (beta-zero):** Y-intercept (value of y when x=0)
- **β_1 (beta-one):** Slope (change in y for unit change in x)
- **x:** Independent variable
- **ε (epsilon):** Error term (residual)

3.2 Geometric Interpretation

Slope (β_1):

- Represents the rate of change
- $\beta_1 = \Delta y / \Delta x$
- Positive slope: y increases as x increases
- Negative slope: y decreases as x increases
- Zero slope: no relationship

Intercept (β_0):

- Where the line crosses the y-axis
- Value of y when all x values are zero
- Not always meaningful in real-world context

3.3 Finding the Best Fit Line

The main idea behind the linear regression model is to fit a line that is the best fit for the data. For this, we use a technique called the **Least Squares Method**.

The Least Squares Method

In Simple Terms: The least squares method is the process of fitting the best curve for a set of data points by reducing the distance between the actual value and predicted value (minimizing the sum of squared residuals). The distance between both values is often known as error, variation, or variance.

Why "Squares"? We square the errors because:

1. Positive and negative errors don't cancel out
2. Large errors are penalized more heavily
3. The math becomes differentiable (we can use calculus)

Understanding the Components

Three Important Quantities:

1. Sum of Squared Errors (SSE) - Unexplained Variance

The distance between the actual value (Y_i) and the predicted value ($\hat{Y}_i$) is the error for that point. When we square these errors and add them up, we get SSE.

$$SSE = \sum (Y_i - \hat{Y}_i)^2$$

What it represents:

- The variation that the model **cannot** explain
- The "leftover" error after fitting the model
- **Must be minimized** to get the best accuracy
- Also called "residual sum of squares"

Visualization: For each data point, there's a vertical line from the point to the regression line. SSE is the sum of the squares of all these vertical distances.

2. Sum of Squared Regression (SSR) - Explained Variance

The distance between the predicted value ($\hat{Y}_i$) and the mean of the dependent variable ($\bar{y}$) represents how much better our model is than just using the mean.

$$SSR = \sum (\hat{Y}_i - \bar{y})^2$$

What it represents:

- The variation that the model **does** explain

- How far predictions deviate from the mean
- **Should be maximized** for better model performance
- Also called "explained sum of squares"

3. Sum of Squared Total (SST) - Total Variance

The total variation in the dependent variable, measured from the mean.

$$SST = \sum (Y_i - \bar{y})^2$$

What it represents:

- Total variation in the data
- The baseline variation before any modeling

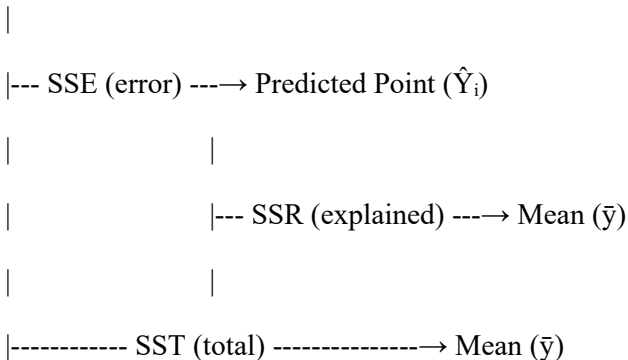
The Fundamental Relationship:

$$SST = SSR + SSE$$

Total Variance = Explained Variance + Unexplained Variance

Visual Understanding:

Actual Point (Y_i)



Advantages and Disadvantages of Least Squares

Advantages:

1. **Differentiable:** Can use calculus to find optimal solution
2. **Closed-form solution:** Can solve directly with the normal equation
3. **Unique solution:** Always gives one best-fit line (if data is not degenerate)

4. **Mathematically elegant:** Well-studied with strong theoretical foundation

Disadvantages:

1. **Not robust to outliers:** Squaring magnifies the effect of outliers
2. **Assumes normal errors:** Works best when residuals are normally distributed
3. **Sensitive to influential points:** One bad point can dramatically change the line

Method 1: Ordinary Least Squares (OLS)

OLS minimizes the sum of squared residuals:

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 = \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

Closed-Form Solution:

$$\beta_1 = \sum ((x_i - \bar{x})(y_i - \bar{y})) / \sum (x_i - \bar{x})^2$$

$$\beta_1 = \text{Cov}(X, Y) / \text{Var}(X)$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Where:

- $\bar{x}$ = mean of x values
- $\bar{y}$ = mean of y values
- $\text{Cov}(X, Y)$ = covariance between X and Y
- $\text{Var}(X)$ = variance of X

Method 2: Gradient Descent

An iterative optimization algorithm:

1. Initialize β_0 and β_1 randomly
2. Calculate predictions: $\hat{y} = \beta_0 + \beta_1 x$
3. Calculate cost: $J = (1/2n) \sum (y_i - \hat{y}_i)^2$
4. Update parameters:
5. $\beta_0 = \beta_0 - \alpha(1/n) \sum (\hat{y}_i - y_i)$ $\beta_1 = \beta_1 - \alpha(1/n) \sum (\hat{y}_i - y_i) x_i$

6. Repeat until convergence

Where α (alpha) is the learning rate.

3.4 Evaluation Metrics

R-Squared (R^2 or Coefficient of Determination)

Measures the proportion of variance in Y explained by X:

$$R^2 = 1 - (SS_{\text{res}} / SS_{\text{tot}})$$

$$SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2 \text{ (Sum of Squared Residuals)}$$

$$SS_{\text{tot}} = \sum (y_i - \bar{y})^2 \text{ (Total Sum of Squares)}$$

Interpretation:

- $R^2 = 1$: Perfect fit (100% variance explained)
- $R^2 = 0.7$: 70% variance explained
- $R^2 = 0$: Model explains nothing
- R^2 can be negative for very poor models

Mean Absolute Error (MAE)

Average absolute difference between predictions and actual values:

$$MAE = (1/n) \sum |y_i - \hat{y}_i|$$

Advantages:

- Easy to interpret (same units as target)
- Less sensitive to outliers than MSE

Mean Squared Error (MSE)

Average of squared differences:

$$MSE = (1/n) \sum (y_i - \hat{y}_i)^2$$

Advantages:

- Penalizes large errors more heavily
- Mathematically convenient (differentiable)

Root Mean Squared Error (RMSE)

Square root of MSE:

$$\text{RMSE} = \sqrt{[1/n]\Sigma(y_i - \hat{y}_i)^2}$$

Advantages:

- Same units as target variable
- Interpretable like MAE but penalizes outliers

3.5 Statistical Inference

Hypothesis Testing for Coefficients

Null Hypothesis: $\beta_1 = 0$ (no relationship)

Alternative Hypothesis: $\beta_1 \neq 0$ (relationship exists)

T-Statistic:

$$t = \beta_1 / \text{SE}(\beta_1)$$

Where $\text{SE}(\beta_1)$ is the standard error of β_1 .

P-value: Probability of observing the data if null hypothesis is true

- $p < 0.05$: Reject null (statistically significant)
- $p \geq 0.05$: Fail to reject null

Confidence Intervals

95% confidence interval for β_1 :

$$\beta_1 \pm (t_{0.025} \times \text{SE}(\beta_1))$$

If the interval doesn't contain 0, the relationship is significant.

3.6 Worked Example: Restaurant Tips

Let's work through a complete real-world example: predicting tip amount based on bill amount.

Problem: You want to know to what degree the tip amount can be predicted by the total bill. The tip is the dependent variable (response variable) and the bill is the independent variable (predictor variable).

Data:

Bill (\$): [50, 68, 80, 90, 110]

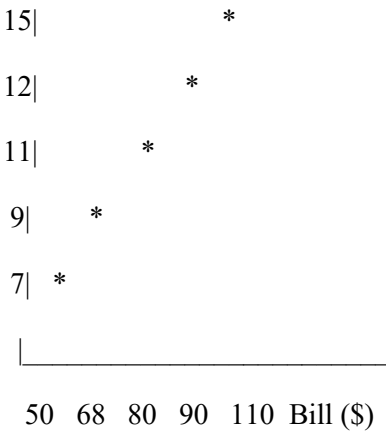
Tip (\$): [7, 9, 11, 12, 15]

Step 1: Check if there is a linear relationship

Before applying linear regression, make a scatter plot to see if there's a relationship between the variables. Remember that **the best fit line will always pass through the centroid**, which is the intersection of $\bar{x}$ (mean of bills) and $\bar{y}$ (mean of tips).

Visual Check:

Tip (\$)



Observation: You can see that there is a **positive relationship**. As the bill amount increases, there is an increase in the tip amount too. Hence, you can use linear regression to predict the response variable.

Step 2: Check the correlation

After plotting a scatter plot and knowing what type of relationship it has, calculate the correlation to know the relationship's strength.

Calculations:

$$\bar{x} = (50 + 68 + 80 + 90 + 110) / 5 = 79.6$$

$$\bar{y} = (7 + 9 + 11 + 12 + 15) / 5 = 10.8$$

$$\Sigma(x - \bar{x})(y - \bar{y}) = (50-79.6)(7-10.8) + (68-79.6)(9-10.8) + \dots = 347.2$$

$$\Sigma(x - \bar{x})^2 = (50-79.6)^2 + (68-79.6)^2 + \dots = 1803.2$$

$$\Sigma(y - \bar{y})^2 = (7-10.8)^2 + (9-10.8)^2 + \dots = 67.2$$

$$r = 347.2 / \sqrt{(1803.2 \times 67.2)} = 347.2 / 348.3 \approx 0.997$$

Interpretation: The correlation is **0.997**, which shows that the relationship is **very strong** and positive. This confirms that linear regression is appropriate.

Step 3: Calculate regression coefficients

Now that you know the relationship is positive and very strong, it's time to calculate the equation of the best-fit line: $\hat{Y} = \beta_0 + \beta_1 x$

Finding β_1 (slope):

The coefficient of regression (slope) tells you the change in the dependent variable if you increase the independent variable by 1 unit.

Formula:

$$\beta_1 = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$$

Calculation:

$$\beta_1 = 347.2 / 1803.2 = 0.1925$$

Interpretation: $\beta_1 = 0.1925$ means that if you increase the bill by \$1, the tip amount will increase by approximately \$0.19 (19 cents).

Finding β_0 (intercept):

Formula:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Calculation:

$$\beta_0 = 10.8 - (0.1925 \times 79.6) = 10.8 - 15.32 = -4.52$$

Interpretation: The intercept is -\$4.52. This may not have practical meaning in real life (you can't have a negative tip for a \$0 bill), but mathematically it positions the line correctly. The intercept is simply where the line crosses the y-axis.

Step 4: Write the final equation

$$\text{Tip} = -4.52 + 0.1925 \times \text{Bill}$$

Or more precisely:

$$\hat{Y} = 0.1925x - 4.52$$

Step 5: Make predictions

Prediction 1: What tip should we expect for a \$75 bill?

$$\text{Tip} = 0.1925(75) - 4.52 = 14.44 - 4.52 = \$9.92$$

Prediction 2: What tip should we expect for a \$100 bill?

$$\text{Tip} = 0.1925(100) - 4.52 = 19.25 - 4.52 = \$14.73$$

Step 6: Evaluate the model

Calculate R²:

$$\text{SST} = \Sigma(y - \bar{y})^2 = 67.2$$

$$\text{SSR} = \Sigma(\hat{y} - \bar{y})^2 = 66.85$$

$$\text{SSE} = \Sigma(y - \hat{y})^2 = 0.35$$

$$R^2 = \text{SSR}/\text{SST} = 66.85/67.2 = 0.995$$

Interpretation: $R^2 = 0.995$ means that 99.5% of the variation in tip amounts can be explained by the bill amount. This is an excellent fit!

Important Note About Calculations

Since there are so many calculations involved, in practice you would use Python libraries or statistical software to make the work easier and avoid calculation errors. The next section will show you how to interpret the results of a linear regression model using statistical software output.

3.7 Limitations of Simple Linear Regression

1. **Only captures linear relationships**
2. **Limited to one predictor variable**
3. **Sensitive to outliers**
4. **Assumes all relevant variables are included**
5. **Cannot model complex interactions**

4. Multiple Linear Regression

4.1 Definition and Formula

Multiple Linear Regression extends simple linear regression to model relationships between multiple independent variables and one dependent variable.

The Equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_n x_n + \varepsilon$$

Or in matrix notation:

$$Y = X\beta + \varepsilon$$

Where:

- **Y**: Vector of target values ($n \times 1$)
- **X**: Matrix of features ($n \times (p+1)$), including intercept column
- **β** : Vector of coefficients ($(p+1) \times 1$)
- **ε** : Vector of errors ($n \times 1$)
- **n**: Number of observations
- **p**: Number of features

4.2 Matrix Formulation

Design Matrix X:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \end{bmatrix}$$

$$\begin{bmatrix} 1 & x_{21} & x_{22} & \dots & x_{2p} \end{bmatrix}$$

$$\begin{bmatrix} \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$$\begin{bmatrix} 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

Normal Equation (Closed-Form Solution):

$$\beta = (X^T X)^{-1} X^T Y$$

This gives the optimal coefficients in one step!

Computational Complexity:

- Matrix inversion: $O(p^3)$
- Practical for $p < 10,000$

- Use gradient descent for larger datasets

4.3 Interpreting Coefficients

Each coefficient represents the change in Y for a one-unit change in that variable, **holding all other variables constant**.

Example: House Price Model

$$\text{Price} = 50 + 0.1(\text{Size}) + 20(\text{Bedrooms}) - 5(\text{Age})$$

Interpretation:

- $\beta_0 = 50$: Base price of \$50,000
- $\beta_1 = 0.1$: Each additional sq ft adds \$100, holding bedrooms and age constant
- $\beta_2 = 20$: Each additional bedroom adds \$20,000, holding size and age constant
- $\beta_3 = -5$: Each year of age reduces price by \$5,000, holding size and bedrooms constant

4.4 Feature Selection

Not all features are equally important. We need methods to select the best subset.

4.4.1 Forward Selection

1. Start with no features
2. Add features one at a time
3. Keep the feature that improves the model most
4. Repeat until no improvement

4.4.2 Backward Elimination

1. Start with all features
2. Remove the least significant feature
3. Refit the model
4. Repeat until all remaining features are significant

4.4.3 Stepwise Selection

Combination of forward and backward:

- Add and remove features at each step
- More thorough but computationally expensive

4.4.4 Regularization Methods

Lasso (L1 Regularization):

$$\text{Cost} = \text{MSE} + \lambda \sum |\beta_j|$$

- Drives some coefficients to exactly zero
- Performs automatic feature selection
- Good for high-dimensional data

Ridge (L2 Regularization):

$$\text{Cost} = \text{MSE} + \lambda \sum \beta_j^2$$

- Shrinks coefficients toward zero
- Keeps all features
- Good when features are correlated

Elastic Net:

$$\text{Cost} = \text{MSE} + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

- Combines L1 and L2
- Balanced approach

4.5 Multicollinearity

Definition: High correlation between independent variables.

Problems:

- Unstable coefficient estimates
- Large standard errors
- Difficulty interpreting individual effects
- Overfitting

Detection:

Variance Inflation Factor (VIF):

$$\text{VIF} = 1 / (1 - R_j^2)$$

Where R_j^2 is the R^2 from regressing feature j on all other features.

Interpretation:

- VIF = 1: No correlation
- VIF < 5: Acceptable
- VIF > 10: Problematic multicollinearity

Solutions:

1. Remove highly correlated features
2. Combine correlated features (PCA)
3. Use regularization (Ridge regression)
4. Collect more data

4.6 Evaluation Metrics for Multiple Regression**Adjusted R-Squared**

Regular R^2 always increases when adding features, even if they're not useful. Adjusted R^2 penalizes for adding unnecessary features:

$$R^2_{\text{adj}} = 1 - [(1 - R^2)(n - 1) / (n - p - 1)]$$

Where:

- n = number of observations
- p = number of features

Use Cases:

- Comparing models with different numbers of features
- Adjusted R^2 can decrease if added feature doesn't improve fit enough

Akaike Information Criterion (AIC)

Measures model quality considering both fit and complexity:

$$AIC = 2p - 2\ln(L)$$

Where:

- p = number of parameters
- L = likelihood of the model

Lower AIC = Better model

Bayesian Information Criterion (BIC)

Similar to AIC but with stronger penalty for complexity:

$$\text{BIC} = \ln(n)p - 2\ln(L)$$

Lower BIC = Better model

4.7 Assumptions and Diagnostics

Checking Linearity

Residual Plot: Plot residuals vs. fitted values

- Should show random scatter
- Patterns indicate non-linearity

Checking Normality

Q-Q Plot: Quantile-Quantile plot of residuals

- Points should follow diagonal line
- Deviations indicate non-normality

Shapiro-Wilk Test:

- $p > 0.05$: Residuals are normal
- $p < 0.05$: Residuals are not normal

Checking Homoscedasticity

Residual Plot: Should show constant spread

- Funnel shape indicates heteroscedasticity

Breusch-Pagan Test:

- Tests for constant variance
- $p < 0.05$: Variance is not constant

Checking for Outliers and Influential Points

Leverage: Measures how far an observation is from the mean **Cook's Distance:** Measures influence of each point

- Cook's $D > 1$: Highly influential point

4.8 Worked Example

Problem: Predict student test score based on study hours, sleep hours, and previous score.

Data (5 students):

Study Hours (X_1): [2, 4, 6, 8, 10]

Sleep Hours (X_2): [6, 7, 8, 7, 9]

Previous Score (X_3): [60, 65, 70, 75, 80]

Test Score (Y): [65, 72, 78, 85, 92]

Step 1: Set up matrices

$X = \begin{bmatrix} 1 & 2 & 6 & 60 \\ 1 & 4 & 7 & 65 \\ 1 & 6 & 8 & 70 \\ 1 & 8 & 7 & 75 \\ 1 & 10 & 9 & 80 \end{bmatrix}$

$Y = \begin{bmatrix} 65 \\ 72 \\ 78 \\ 85 \\ 92 \end{bmatrix}$

Step 2: Calculate $\beta = (X^T X)^{-1} X^T Y$

(Using software/calculator for matrix operations)

Result:

$\beta_0 = 10.5$

$\beta_1 = 2.3$ (Study Hours)

$\beta_2 = 1.8$ (Sleep Hours)

$\beta_3 = 0.4$ (Previous Score)

Final Model:

$$\text{Score} = 10.5 + 2.3(\text{Study}) + 1.8(\text{Sleep}) + 0.4(\text{Previous})$$

Interpretation:

- Each study hour adds 2.3 points
- Each sleep hour adds 1.8 points
- Each point on previous test adds 0.4 points

Prediction for new student:

- Study: 7 hours, Sleep: 8 hours, Previous: 72
- $\text{Score} = 10.5 + 2.3(7) + 1.8(8) + 0.4(72) = 10.5 + 16.1 + 14.4 + 28.8 = 69.8 \approx 70$

4.9 Common Pitfalls

1. **Including too many features (overfitting)**
2. **Ignoring multicollinearity**
3. **Not checking assumptions**
4. **Using R^2 alone for model selection**
5. **Extrapolating beyond data range**
6. **Confusing correlation with causation**

5. Polynomial Linear Regression

5.1 Definition and Motivation

Polynomial regression models non-linear relationships by adding polynomial terms while remaining "linear" in the coefficients.

Why Polynomial? Real-world relationships are often non-linear:

- Growth curves (S-shaped)
- Diminishing returns
- Parabolic patterns
- Complex interactions

The Key Insight: It's still called "linear" regression because it's linear in the **parameters** (β), not necessarily in the features (x).

5.2 Mathematical Formulation

Polynomial of Degree d:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_d x^d + \varepsilon$$

Common Forms:

Quadratic (Degree 2):

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

Cubic (Degree 3):

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

Quartic (Degree 4):

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4$$

5.3 Feature Engineering

Polynomial regression is achieved by **creating new features** that are powers of the original feature:

Original feature: $X = [x_1, x_2, x_3, \dots, x_n]$

Transformed features (degree 3):

$$X_{\text{poly}} = [1, x_1, x_1^2, x_1^3]$$

$$[1, x_2, x_2^2, x_2^3]$$

$[1, x_3, x_3^2, x_3^3]$ $[\vdots \vdots \vdots \vdots]$ $[1, x_n, x_n^2, x_n^3]$

Then we apply standard linear regression on X_{poly} !

5.4 Choosing the Degree

Underfitting (Too Low Degree):

- Model too simple
- High bias, low variance
- Poor training and test performance

Overfitting (Too High Degree):

- Model too complex
- Low bias, high variance
- Excellent training but poor test performance

Methods to Choose Degree:

5.4.1 Visual Inspection

Plot the fitted curve and see if it captures the pattern.

5.4.2 Cross-Validation

Split data into train/validation sets and test different degrees:

For degree $d = 1$ to max_degree :

Fit model on training data

Evaluate on validation data

Track error

Choose degree with lowest validation error

5.4.3 Information Criteria

- Use AIC or BIC
- Balance fit quality with model complexity

5.4.4 Learning Curves

Plot training and validation error vs. degree:

- Sweet spot is where curves are close and errors are low

5.5 Multiple Features with Polynomial Terms

When you have multiple features, polynomial regression can include:

- Individual polynomial terms: x_1^2, x_2^3
- Interaction terms: $x_1x_2, x_1^2x_2$

Example with 2 features (degree 2):

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1^2 + \beta_4x_2^2 + \beta_5x_1x_2$$

Number of terms grows quickly: For p features and degree d , the number of terms is:

$$C(p + d, d) = (p + d)! / (p! \times d!)$$

Example: 3 features, degree 3 $\rightarrow$ 20 terms!

5.6 Regularization for Polynomial Regression

High-degree polynomials create many features $\rightarrow$ high risk of overfitting.

Solution: Use regularization

Ridge Polynomial Regression:

$$\text{Cost} = \text{MSE} + \lambda \sum \beta_j^2$$

- Shrinks coefficients of high-degree terms
- Reduces overfitting without removing features

Recommended λ values:

- Start with $\lambda = 0.1, 1, 10, 100$
- Use cross-validation to find optimal λ

5.7 Advantages and Disadvantages

Advantages:

1. Captures non-linear relationships
2. Simple extension of linear regression

3. No need for complex algorithms
4. Interpretable for low degrees
5. Flexible curve fitting

Disadvantages:

1. Very sensitive to outliers
2. Overfitting risk with high degrees
3. Poor extrapolation beyond data range
4. Multicollinearity among polynomial terms
5. Loss of interpretability for high degrees

5.8 Worked Example

Problem: Model the trajectory of a thrown ball.

Physics: Height follows a parabolic path (quadratic relationship)

Data:

Time (seconds): [0, 1, 2, 3, 4, 5]

Height (meters): [0, 45, 70, 75, 60, 25]

Step 1: Try Linear Regression

$$\text{Height} = \beta_0 + \beta_1(\text{Time})$$

Result: Poor fit ($R^2 \approx 0.5$)

Step 2: Try Quadratic Regression

$$\text{Height} = \beta_0 + \beta_1(\text{Time}) + \beta_2(\text{Time}^2)$$

Create polynomial features:

Time²: [0, 1, 4, 9, 16, 25]

Fit using matrix methods:

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \end{bmatrix}$$

[1 3 9]

[1 4 16]

[1 5 25]

$Y = [0, 45, 70, 75, 60, 25]^T$

Result:

$\beta_0 = 0$

$\beta_1 = 50$

$\beta_2 = -5$

Final Model:

Height = 50(Time) - 5(Time²)

Evaluation:

- $R^2 \approx 0.98$ (excellent fit!)
- Matches physical parabolic trajectory

Prediction at t = 2.5 seconds:

Height = 50(2.5) - 5(2.5²)

Height = 125 - 31.25 = 93.75 meters

Physical Interpretation:

- Initial velocity: 50 m/s
- Acceleration: -10 m/s² (gravity, approximately)

5.9 Best Practices

1. **Start simple:** Try linear before polynomial
2. **Normalize features:** Prevent numerical instability
3. **Use cross-validation:** Choose degree objectively
4. **Apply regularization:** Especially for degree > 3
5. **Visualize results:** Plot fitted curve vs. data

6. **Check residuals:** Ensure assumptions are met
7. **Avoid high degrees:** Rarely need degree > 5
8. **Be cautious with extrapolation:** Polynomials behave badly outside data range

5.10 Alternatives to Polynomial Regression

When polynomial regression doesn't work well:

1. **Spline Regression:** Piecewise polynomials (more flexible)
2. **Generalized Additive Models (GAM):** Non-parametric smoothing
3. **Decision Trees/Random Forests:** Handle non-linearity naturally
4. **Neural Networks:** Universal function approximators
5. **Support Vector Regression:** With non-linear kernels

6. Advanced Topics

6.1 Robust Regression

Standard linear regression is sensitive to outliers. Robust methods reduce this sensitivity.

Huber Regression

Uses a loss function that's quadratic for small errors and linear for large errors:

$$\text{Loss} = \begin{cases} \frac{1}{2}e^2 & \text{for } |e| \leq \delta \\ \delta|e| - \frac{1}{2}\delta^2 & \text{for } |e| > \delta \end{cases}$$

RANSAC (Random Sample Consensus)

1. Randomly sample subset of data
2. Fit model to subset
3. Count inliers (points close to model)
4. Repeat and keep best model

6.2 Weighted Least Squares

When errors have different variances (heteroscedasticity), use weights:

$$\text{Cost} = \sum w_i (y_i - \hat{y}_i)^2$$

Where $w_i = 1/\text{variance of observation } i$

6.3 Quantile Regression

Predicts conditional quantiles instead of conditional mean:

Median Regression (50th percentile): More robust to outliers than mean regression.

90th Percentile Regression: Useful for risk assessment and worst-case scenarios.

6.4 Bayesian Linear Regression

Treats parameters as random variables with probability distributions:

Advantages:

- Provides uncertainty estimates
- Incorporates prior knowledge
- Natural regularization

- Handles small datasets well

Posterior Distribution:

$$P(\beta|\text{Data}) \propto P(\text{Data}|\beta) \times P(\beta)$$

6.5 Time Series Regression

When data has temporal ordering:

Autoregressive Models:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \dots + \varepsilon$$

Considerations:

- Autocorrelation in residuals
- Stationarity requirements
- Trend and seasonality

6.6 Non-parametric Regression

Kernel Regression: Weighted average of nearby points.

Locally Weighted Regression (LOESS): Fits local polynomial at each point.

Advantages:

- No assumptions about functional form
- Flexible

Disadvantages:

- Computationally expensive
- Hard to interpret
- Poor extrapolation

6.7 Feature Scaling and Transformation

Standardization (Z-score normalization)

$$x' = (x - \mu) / \sigma$$

Centers data at 0 with standard deviation 1.

Min-Max Scaling

$$x' = (x - \min) / (\max - \min)$$

Scales to [0, 1] range.

Log Transformation

$$x' = \log(x)$$

For right-skewed data.

Box-Cox Transformation

Automatically finds best power transformation:

$$x' = (x^\lambda - 1) / \lambda \quad \text{if } \lambda \neq 0$$

$$x' = \log(x) \quad \text{if } \lambda = 0$$

6.8 Handling Categorical Variables

One-Hot Encoding

Convert categories to binary variables:

Color: [Red, Blue, Red, Green]

→

Red: [1, 0, 1, 0]

Blue: [0, 1, 0, 0]

Green: [0, 0, 0, 1]

Note: Drop one category to avoid multicollinearity (dummy variable trap).

Label Encoding

Assign integers to categories:

Color: [Red, Blue, Green] → [0, 1, 2]

Warning: Implies ordinal relationship. Only use for truly ordinal data.

6.9 Interaction Terms

Model how the effect of one variable depends on another:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2)$$

Example: Effect of advertising on sales might depend on price.

Interpretation: The coefficient β_3 shows how the effect of x_1 changes with x_2 .

7. Practical Implementation

7.1 Python Implementation with scikit-learn

Simple Linear Regression

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.metrics import r2_score, mean_squared_error


# Sample data

X = np.array([[1], [2], [3], [4], [5]])

y = np.array([2, 4, 5, 4, 5])


# Create and fit model

model = LinearRegression()

model.fit(X, y)


# Get coefficients

print(f"Intercept: {model.intercept_}")

print(f"Slope: {model.coef_[0]}")


# Make predictions

y_pred = model.predict(X)


# Evaluate

r2 = r2_score(y, y_pred)

mse = mean_squared_error(y, y_pred)
```

```
print(f'R²: {r2:.4f}')
print(f'MSE: {mse:.4f}')

# Visualize
plt.scatter(X, y, color='blue', label='Actual')
plt.plot(X, y_pred, color='red', label='Predicted')
plt.xlabel('X')
plt.ylabel('y')
plt.legend()
plt.show()
```

Multiple Linear Regression

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

```
# Sample data (multiple features)
X = np.array([[1, 2], [2, 3], [3, 4], [4, 5], [5, 6]])
y = np.array([3, 5, 7, 9, 11])
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Scale features (optional but recommended)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

```
# Create and fit model
```

```
model = LinearRegression()
```

```
model.fit(X_train_scaled, y_train)
```

```
# Predictions
```

```
y_pred = model.predict(X_test_scaled)
```

```
# Coefficients
```

```
print(f"Intercept: {model.intercept_}")
```

```
print(f"Coefficients: {model.coef_}")
```

```
# Evaluate
```

```
print(f"Training R2: {model.score(X_train_scaled, y_train):.4f}")
```

```
print(f"Test R2: {model.score(X_test_scaled, y_test):.4f}")
```

Polynomial Regression

```
from sklearn.preprocessing import PolynomialFeatures
```

```
from sklearn.pipeline import make_pipeline
```

```
# Sample data
```

```
X = np.array([[1], [2], [3], [4], [5]])
```

```
y = np.array([1, 4, 9, 16, 25]) # Quadratic relationship
```

```
# Create polynomial features and fit in one pipeline
```

```
degree = 2
```

```
poly_model = make_pipeline(  
    PolynomialFeatures(degree),  
    LinearRegression()  
)
```

```
poly_model.fit(X, y)
```

```
# Predictions
```

```
y_pred = poly_model.predict(X)
```

```
# Visualize
```

```
X_plot = np.linspace(0, 6, 100).reshape(-1, 1)
```

```
y_plot = poly_model.predict(X_plot)
```

```
plt.scatter(X, y, color='blue', label='Actual')
```

```
plt.plot(X_plot, y_plot, color='red', label=f'Degree {degree}')
```

```
plt.xlabel('X')
```

```
plt.ylabel('y')
```

```
plt.legend()
```

```
plt.show()
```

```
# R2 score
```

```
print(f'R2: {r2_score(y, y_pred):.4f}')
```

Ridge and Lasso Regression

```
from sklearn.linear_model import Ridge, Lasso
```

```
from sklearn.model_selection import GridSearchCV
```

```

# Sample data

X = np.random.randn(100, 10)
y = np.random.randn(100)


# Ridge Regression

ridge = Ridge(alpha=1.0)
ridge.fit(X, y)


# Lasso Regression

lasso = Lasso(alpha=0.1)
lasso.fit(X, y)


# Hyperparameter tuning with cross-validation

param_grid = {'alpha': [0.001, 0.01, 0.1, 1, 10, 100]}

ridge_cv = GridSearchCV(Ridge(), param_grid, cv=5)
ridge_cv.fit(X, y)

print(f"Best alpha: {ridge_cv.best_params_['alpha']}")
print(f"Best score: {ridge_cv.best_score_:.4f}")

```

7.2 Model Diagnostics in Python

```

import scipy.stats as stats


# After fitting a model...

residuals = y - y_pred

```



```
# 1. Residual plot
```

```
plt.figure(figsize=(10, 4))
```

```
plt.subplot(1, 2, 1)
```

```
plt.scatter(y_pred, residuals)
```

```
plt.axhline(y=0, color='r', linestyle='--')
```

```
plt.xlabel('Fitted values')
```

```
plt.ylabel('Residuals')
```

```
plt.title('Residual Plot')
```

```
# 2. Q-Q plot
```

```
plt.subplot(1, 2, 2)
```

```
stats.probplot(residuals, dist="norm", plot=plt)
```

```
plt.title('Q-Q Plot')
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# 3. Check for normality
```

```
_, p_value = stats.shapiro(residuals)
```

```
print(f"Shapiro-Wilk test p-value: {p_value:.4f}")
```

```
if p_value > 0.05:
```

```
    print("Residuals appear normally distributed")
```

```
else:
```

```
    print("Residuals may not be normally distributed")
```

7.3 Interpreting Linear Regression Output

When you run a linear regression model using statistical software (like Python's statsmodels), you get a detailed summary output. Understanding this output is crucial for properly interpreting your model. Let's break down each component.

Getting the Summary in Python

```
import statsmodels.api as sm

# Add constant term for intercept
X_with_const = sm.add_constant(X)

# Fit the model
model = sm.OLS(y, X_with_const).fit()

# Display comprehensive summary
print(model.summary())
```

Understanding the Output Components

PART 1: Model Overview

R-squared (Coefficient of Determination)

R-squared states the proportion of the variability in the dependent variable that is explained by the model, i.e., what percent of the real-world variation the model represents.

$$R^2 = 1 - (SSE / SST) = SSR / SST$$

Real-World Example: You have an independent variable as "CGPA" and a dependent variable as "Job Package". You want to apply to XYZ college and know the package you will receive after graduating.

- **Without CGPA info:** The college tells you the average package of previous students (using the mean $\bar{y}$).
- **With CGPA info:** The college can predict your package using linear regression based on past data.

The R^2 score shows the superior results that linear regression can give instead of just basing predictions on the mean.

Can R^2 be 0? Yes, if both the regression line and mean line have the same error ($SSE = SST$). If the regression line overlaps with the mean line, you can say that both have the same error. This means providing your CGPA has no importance because you will get the same result even without that information.

Can R^2 be 1? Yes. For this to happen, you need $SSE = 0$. This occurs when the regression line perfectly fits all the points, which is very rare in real-world data. It means the model explains 100% of the variance.

Can R^2 be negative? Yes, but only when $SSE > SST$. This means the regression line error is more than just using the mean (worst-case scenario). This can happen only when the regression line is terrible and performs worse than simply predicting the mean for every observation.

Interpretation Guidelines:

- **$R^2 > 0.80$ (80%):** The model is a **good fit**
- **$0.60 < R^2 < 0.80$ (60-80%):** The model is an **okay fit**
- **$R^2 < 0.60$ (below 60%):** The model **needs improvement**

If your R^2 value is low, you may need to:

- Check your independent variables again
- Look for outliers
- Consider adding more relevant features
- Try polynomial or interaction terms

Adjusted R-squared

Every time you add a new input variable, R^2 will increase (or stay the same), even if the variable is completely irrelevant. To address this, we use **Adjusted R^2** , which is a modified version that penalizes for adding unnecessary features.

$$R^2_{\text{adj}} = 1 - [(1 - R^2)(n - 1) / (n - p - 1)]$$

Where:

- n = number of observations
- p = number of independent variables

Key Points:

- Adjusted R^2 is **always lower** than R^2
- If you add variables that don't affect the target, adjusted R^2 will **decrease** while R^2 increases
- Use adjusted R^2 when comparing models with different numbers of features

When to worry: Usually, the values of R^2 and adjusted R^2 are similar. But if you see a **large difference**, you need to check your independent variables and see if there is any real relationship between the target variable and the independent variables. A big gap suggests you have too many irrelevant features.

PART 2: ANOVA - F-statistic

F-statistic reveals whether at least one of the coefficients in the model is statistically different from zero. In other words, it tests if the overall model is useful.

Hypotheses:

- **Null Hypothesis (H_0):** The model with no independent variables fits the data as well as your model (all β coefficients = 0)
- **Alternative Hypothesis (H_1):** Your model fits the data better than the intercept-only model (at least one $\beta \neq 0$)

Using P-value to Test: The p-value associated with the F-statistic helps check whether there is enough evidence to accept the alternative hypothesis.

If you are testing at a **95% confidence interval**:

- **p-value < 0.05** (level of significance): Reject $H_0 \rightarrow$ The model is statistically significant
- **p-value > 0.05:** Fail to reject $H_0 \rightarrow$ The model is not statistically significant

Interpretation: A significant F-test means your model as a whole is meaningful and that at least some of your features have a real relationship with the target variable.

PART 3: Variables Table - Understanding Coefficients

This is where you examine each individual feature's contribution to the model.

Coefficient

The coefficient of a variable shows that if you increase the independent variable by **1 unit**, holding other variables constant (remember, linear regression assumes no multicollinearity), then the dependent variable will change by that coefficient value.

Example Interpretation: If the coefficient of "age" is **257.4050**:

- Increasing age by 1 year will increase the target variable by 257.4050 units
- If the coefficient was negative (e.g., -353.4491 for "region"), increasing region by 1 unit would decrease the target variable by 353.4491 units

Coefficient Near Zero: If the coefficient value is close to 0, you can say that there is **no relationship** between that variable and the target variable. This feature may not be useful for your model.

Standard Error (Std. Error)

To understand standard error, you first need to know about standard deviation.

Standard Deviation tells the variation of values from the mean or how spread the data is. About 95% of values lie within 2 standard deviations of the mean.

Standard Error in Simple Terms: Imagine you have a population and you pick 10 different samples from it. If you find the mean of each sample and plot them on a standard normal graph, the standard deviation of these sample means is called the **standard error**.

In Regression: Standard error gives the estimated standard deviation of the distribution of coefficients.

What does this mean? For every 1 unit increase in age, the target variable goes up by 257.4050. If you re-run the model with different samples, there may be some variation in this coefficient. The standard error shows how much the coefficient might vary if you re-run the model.

Example: If age has a standard error of **11.878**, this means the coefficient of age may vary by approximately 11.878 if you re-run the model with different data samples.

t-statistic (t-value)

The t-statistic is calculated by dividing the coefficient by its standard error:

$$\text{t-statistic} = \text{Coefficient} / \text{Std. Error}$$

Purpose: The t-statistic helps determine if the coefficient value is real or if it occurred by chance.

Hypotheses:

- **Null Hypothesis (H_0):** The coefficient at the population level is 0 (no relationship)
- **Alternative Hypothesis (H_1):** The coefficient is not 0 (relationship exists)

Decision Rule: If the absolute value of the t-statistic is greater than the critical value from the t-distribution table (typically around 2 for 95% confidence), you reject the null hypothesis.

Interpretation:

- **Higher magnitude of t-statistic** = more significant the variable
- The further the t-value is from 0 (in either direction), the more confident we are that the coefficient is not zero

$P > |t|$ (P-value for Individual Coefficients)

The p-value is the **probability that the null hypothesis is true**. It shows how likely it is that the relationship you're seeing is just due to random chance.

Example: If you look at the p-value of age and it's approximately **0.000**, this means the probability that there's no relationship between age and the target variable is essentially zero. In other words, it's extremely unlikely that the coefficient is actually zero.

Decision Rule: Using a **5% significance level** ($\alpha = 0.05$):

- **p-value < 0.05:** Reject the null hypothesis → There IS a significant relationship between the variable and target
- **p-value > 0.05:** Fail to reject the null hypothesis → There is NO significant relationship

Important Relationship:

- **Higher t-statistic → Lower p-value → Higher significance**
- The variable's effect is real and didn't happen by chance

Typical Significance Levels:

- **p < 0.001:** Highly significant (***)
- **p < 0.01:** Very significant (**)
- **p < 0.05:** Significant (*)
- **p > 0.05:** Not significant

PART 4: Confidence Intervals

Most software also provides confidence intervals for coefficients (e.g., 95% CI).

Interpretation: We are 95% confident that the true population coefficient lies within this interval.

Example: If the 95% CI for the age coefficient is [234.1, 280.7]:

- We're 95% confident the true effect of age is between 234.1 and 280.7
- If the interval doesn't include 0, the variable is significant

Summary of Interpretation Process

Step 1: Check R^2 and Adjusted R^2

- Is the model explaining a good proportion of variance?
- Are there too many irrelevant features?

Step 2: Check F-statistic and its p-value

- Is the overall model statistically significant?

Step 3: For each variable, check:

- **Coefficient:** What's the effect size and direction?
- **Standard Error:** How variable is this estimate?
- **t-statistic:** How many standard errors away from zero?

- **p-value:** Is this variable statistically significant?

Step 4: Make decisions

- Remove variables with high p-values (> 0.05)
- Focus on variables with large, significant coefficients
- Check if results make practical/business sense

Congratulations! Understanding how to interpret linear regression output is crucial for making informed decisions based on your model. It's always better to know what's happening under the hood so you can proceed confidently and make good predictions!

7.4 Feature Selection Example

```
from sklearn.feature_selection import f_regression, SelectKBest
```

```
# Feature selection using F-statistic
```

```
selector = SelectKBest(f_regression, k=3)
```

```
X_selected = selector.fit_transform(X, y)
```

```
# Get selected feature indices
```

```
selected_indices = selector.get_support(indices=True)
```

```
print(f"Selected features: {selected_indices}")
```

```
# Get F-scores and p-values
```

```
f_scores, p_values = f_regression(X, y)
```

```
for i, (score, p_val) in enumerate(zip(f_scores, p_values)):
```

```
    print(f"Feature {i}: F-score = {score:.2f}, p-value = {p_val:.4f}")
```

8. Real-World Applications

8.1 Business and Finance

Example 1: Advertising Spend and Revenue

Businesses frequently use linear regression to comprehend the connection between advertising spending and revenue. For instance, they might apply the linear regression model using advertising spend as an independent variable or predictor variable and revenue as the response variable.

The Model:

$$\text{revenue} = \beta_0 + \beta_1(\text{ad spending})$$

Real-World Application:

- β_0 : Base revenue without advertising
- β_1 : Additional revenue generated per dollar spent on advertising
- Companies use this to optimize advertising budgets and predict ROI

Example Interpretation: If $\beta_1 = 3.5$, then for every \$1,000 spent on advertising, revenue increases by \$3,500.

Other Business Applications

Stock Price Prediction:

- Model stock returns based on market factors
- Type: Multiple regression with time-series considerations
- Features: Market index, sector performance, company fundamentals

Credit Scoring:

- Predict default probability
- Type: Logistic regression (cousin of linear regression)
- Features: Income, debt, credit history, employment status

8.2 Healthcare

Example 2: Drug Dosage and Blood Pressure

Linear regression may be used in the medical field to understand the relationships between drug dosage and patient blood pressure. Researchers may administer different measurements of a specific medication to patients and see how their blood pressure responds.

The Model:

$$\text{blood pressure} = \beta_0 + \beta_1(\text{dosage})$$

Real-World Application:

- β_0 : Baseline blood pressure without medication
- β_1 : Change in blood pressure per mg of drug
- Doctors use this to determine optimal dosage for patients
- Helps understand dose-response relationships

Example Interpretation: If $\beta_1 = -2.5$, then each additional mg of the drug reduces blood pressure by 2.5 mmHg.

Safety Considerations: Researchers must ensure the relationship remains linear within therapeutic ranges, as very high doses may have non-linear (potentially dangerous) effects.

Other Healthcare Applications

Disease Progression:

- Predict patient outcomes based on clinical measurements
- Type: Multiple linear regression
- Features: Age, BMI, blood pressure, lab results

Hospital Stay Duration:

- Predict length of hospital stay
- Type: Polynomial regression (may follow non-linear pattern)
- Features: Diagnosis, severity, age, comorbidities

8.3 Environmental Science and Agriculture

Example 3: Rainfall, Fertilizer, and Crop Yield

Agriculture scientists frequently use linear regression to see the impact of rainfall and fertilizer on the amount of fruits/vegetables yielded. For instance, scientists might use different amounts of fertilizer and observe the effect of rain on different fields to ascertain how it affects crop yield.

The Model (Multiple Linear Regression):

$$\text{crop yield} = \beta_0 + \beta_1(\text{rainfall}) + \beta_2(\text{fertilizer})$$

Real-World Application:

- β_0 : Base crop yield with no rainfall or fertilizer
- β_1 : Change in yield per inch of rainfall
- β_2 : Change in yield per kg of fertilizer
- Farmers use this to optimize resource allocation
- Helps predict yields under different weather scenarios

Example Interpretation:

- If $\beta_1 = 50$ and $\beta_2 = 30$, then:
 - Each additional inch of rainfall increases yield by 50 kg/hectare
 - Each additional kg of fertilizer increases yield by 30 kg/hectare

Planning Application: Farmers can decide whether investing in irrigation (to compensate for low rainfall) is more cost-effective than using additional fertilizer.

Other Environmental Applications

Climate Modeling:

- Predict temperature changes
- Type: Multiple regression with polynomial terms
- Features: CO₂ levels, solar radiation, ocean currents

Air Quality Forecasting:

- Predict pollution levels
- Type: Time-series regression
- Features: Traffic, weather, industrial activity

8.4 Social Sciences

Education:

- Predict student performance
- Type: Multiple regression
- Features: Study hours, attendance, previous grades, socioeconomic factors

Economics:

- Demand forecasting

- Type: Polynomial regression (diminishing returns)
- Features: Price, income, competitor pricing

Real Estate:

- House price prediction
- Type: Multiple regression with interactions
- Features: Size, location, age, bedrooms, economic conditions

8.5 Engineering

Manufacturing Quality Control:

- Predict product quality metrics
- Type: Multiple regression
- Features: Temperature, pressure, material properties, machine settings

Energy Consumption:

- Forecast electricity demand
- Type: Multiple regression with polynomial and seasonal terms
- Features: Temperature, day of week, economic activity

Structural Analysis:

- Predict stress and strain in materials
- Type: Polynomial regression
- Features: Load, material properties, geometry

9. Summary and Best Practices

9.1 Quick Reference Guide

Regression Type	Use When	Advantages	Disadvantages
Simple Linear	One predictor, linear relationship	Simple, interpretable	Limited to one variable
Multiple Linear	Multiple predictors, linear relationships	Handles many variables	Multicollinearity issues
Polynomial	Non-linear relationships	Captures curves	Overfitting risk

9.2 Decision Tree: Which Regression to Use?

Start

↓

Do you have one or multiple features?

↓

One feature → Is the relationship linear?

↓

Yes

↓

Simple Linear

Regression

↓

No

↓

Polynomial

Regression

Multiple features → Are relationships linear?

↓

Yes

↓

Multiple Linear

Regression

↓

No

↓

Polynomial with

multiple features

9.3 The Regression Workflow

Step 1: Problem Definition

- What are you trying to predict?
- What data do you have?
- What's the business/research question?

Step 2: Data Preparation

- Collect and clean data
- Handle missing values
- Remove outliers (or use robust methods)
- Split into train/test sets

Step 3: Exploratory Analysis

- Visualize relationships
- Check correlations
- Identify potential features

Step 4: Feature Engineering

- Create polynomial features if needed
- Create interaction terms
- Transform variables (log, sqrt, etc.)
- Encode categorical variables

Step 5: Model Training

- Start simple (linear)
- Add complexity if needed
- Use cross-validation
- Try regularization

Step 6: Evaluation

- Check R^2 , MSE, RMSE

- Examine residual plots
- Verify assumptions
- Compare train/test performance

Step 7: Interpretation

- Understand coefficients
- Identify important features
- Check statistical significance

Step 8: Deployment

- Document model
- Create prediction pipeline
- Monitor performance
- Update as needed

9.4 Common Mistakes to Avoid

1. Overfitting

- Using too many features
- Too high polynomial degree
- Solution: Cross-validation, regularization

2. Underfitting

- Model too simple for the data
- Solution: Add features, increase complexity

3. Ignoring Assumptions

- Not checking linearity, normality, etc.
- Solution: Always perform diagnostics

4. Data Leakage

- Using future information to predict past
- Solution: Careful train/test splitting

5. Multicollinearity

- Highly correlated features
- Solution: Remove features, use PCA, or Ridge

6. Extrapolation

- Predicting beyond data range
- Solution: Be cautious, acknowledge uncertainty

7. Ignoring Outliers

- One outlier can drastically affect the fit
- Solution: Investigate, remove if appropriate, or use robust methods

8. Feature Scaling Issues

- Not scaling for regularization
- Solution: Always scale when using Ridge/Lasso

9. Causation vs. Correlation

- Assuming regression shows causation
- Solution: Remember: correlation $\neq$ causation

10. Not Using Validation Set

- Testing on training data
- Solution: Always use separate test set

9.5 Best Practices Checklist

Before Modeling:

- ☐ Understand the problem domain
- ☐ Clean and preprocess data
- ☐ Visualize data and relationships
- ☐ Check for missing values and outliers
- ☐ Split data into train/test sets

During Modeling:

- ☐ Start with simple models
- ☐ Use cross-validation
- ☐ Check for overfitting
- ☐ Try regularization if needed
- ☐ Document experiments

After Modeling:

- ☐ Verify assumptions (residual plots, Q-Q plot)
- ☐ Check for influential points
- ☐ Interpret coefficients carefully
- ☐ Test on holdout set
- ☐ Compare multiple models

For Production:

- ☐ Document model thoroughly
- ☐ Create reproducible pipeline
- ☐ Monitor performance over time
- ☐ Plan for model updates
- ☐ Set up alerts for performance degradation

9.6 Key Takeaways

Simple Linear Regression:

- Foundation of all regression
- One feature, straight line
- Easy to interpret and implement

Multiple Linear Regression:

- Extension to multiple features
- Coefficients show individual effects
- Watch for multicollinearity

Polynomial Regression:

- Captures non-linear patterns
- Still "linear" in parameters
- Risk of overfitting with high degrees

General Principles:

- Simpler is often better
- Always validate assumptions
- Use regularization for complex models
- Interpretability matters
- Domain knowledge is crucial

9.7 Further Learning Resources**Books:**

- "An Introduction to Statistical Learning" by James, Witten, Hastie, Tibshirani
- "The Elements of Statistical Learning" by Hastie, Tibshirani, Friedman
- "Applied Regression Analysis" by Draper and Smith

Online Courses:

- Coursera: Machine Learning by Andrew Ng
- edX: Statistical Learning
- Fast.ai: Practical Deep Learning

Practice Datasets:

- UCI Machine Learning Repository
- Kaggle competitions
- scikit-learn built-in datasets

Tools:

- Python: scikit-learn, statsmodels
- R: lm(), glm(), caret

- Excel: Built-in regression tools (for simple cases)

9.8 Final Thoughts

Linear regression, in its various forms, remains one of the most valuable tools in data science because:

1. **Interpretability:** You can explain what the model does
2. **Speed:** Fast to train and predict
3. **Baseline:** Excellent starting point for any regression problem
4. **Theory:** Strong statistical foundation
5. **Versatility:** Applicable across many domains

Remember: **Start simple, validate thoroughly, and only add complexity when needed.**

The best model is not always the most complex one—it's the one that best balances accuracy, interpretability, and generalization to new data.

Appendix A: Mathematical Derivations

A.1 Derivation of OLS Coefficients

Objective: Minimize Sum of Squared Residuals (SSR)

$$SSR = \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

Take partial derivatives and set to zero:

$$\partial SSR / \partial \beta_0 = -2 \sum (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\partial SSR / \partial \beta_1 = -2 \sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

Solving the first equation:

$$\sum y_i - n\beta_0 - \beta_1 \sum x_i = 0$$

$$\beta_0 = (\sum y_i - \beta_1 \sum x_i) / n$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Substituting into second equation and solving:

$$\sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\sum x_i y_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\sum x_i y_i - \bar{y} \sum x_i + \beta_1 \bar{x} \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\sum x_i y_i - n\bar{x}\bar{y} = \beta_1 (\sum x_i^2 - n\bar{x}^2)$$

$$\beta_1 = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2$$

A.2 Proof that $R^2 = r^2$

For simple linear regression, R^2 equals the square of the correlation coefficient.

Correlation coefficient:

$$r = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\sum (x_i - \bar{x})^2 \times \sum (y_i - \bar{y})^2]}$$

Can be shown that:

$$R^2 = 1 - SSR/SST = r^2$$

Appendix B: Glossary

Adjusted R²: R² modified to penalize for number of features.

Bias: Systematic error in predictions (underfitting).

Coefficient: Parameter in regression equation (β).

Collinearity: Correlation between independent variables.

Cross-Validation: Technique to assess model performance on unseen data.

Dependent Variable: Variable being predicted (Y).

Feature: Input variable (X).

Heteroscedasticity: Non-constant variance of residuals.

Homoscedasticity: Constant variance of residuals.

Independent Variable: Predictor variable (X).

Intercept: Value of Y when all X's are zero (β_0).

Leverage: Measure of how far a point is from the mean.

MSE: Mean Squared Error.

Multicollinearity: High correlation among multiple features.

OLS: Ordinary Least Squares (fitting method).

Outlier: Data point far from others.

Overfitting: Model too complex, poor generalization.

P-value: Probability of observing data if null hypothesis true.

R²: Proportion of variance explained by model.

Regularization: Technique to prevent overfitting.

Residual: Difference between actual and predicted values.

RMSE: Root Mean Squared Error.

Slope: Rate of change (β_1).

Underfitting: Model too simple, high bias.

Variance: Variability in model predictions.

VIF: Variance Inflation Factor (multicollinearity measure).

Appendix C: Practice Problems

Problem 1: Simple Linear Regression

Data on advertising spend and sales:

Ad Spend (\$1000s): [1, 2, 3, 4, 5]

Sales (\$1000s): [10, 12, 15, 17, 20]

a) Calculate β_0 and β_1 by hand. b) Predict sales for \$3,500 ad spend. c) Calculate R^2 .

Problem 2: Multiple Linear Regression

Predict house price based on:

Size (sq ft), Age (years), Bedrooms

[1500, 5, 3] → \$250,000

[2000, 10, 4] → \$300,000

[1800, 3, 3] → \$280,000

a) Set up the design matrix X and vector Y . b) Describe what each coefficient would represent.

Problem 3: Polynomial Regression

Temperature data over time:

Time: [0, 1, 2, 3, 4, 5]

Temp: [20, 25, 28, 29, 28, 25]

a) Why might polynomial regression be appropriate?

b) What degree polynomial would you try first?

c) What would be a sign of overfitting?

This comprehensive guide has taken you from the basics of supervised learning through the three main types of linear regression, advanced topics, and practical implementation. You now have the foundation to apply regression analysis to real-world problems and continue advancing your skills.

Remember: Practice is key! Work through the examples, implement the code, and apply these techniques to your own datasets.

The future of AI is bright, and you can be part of it!